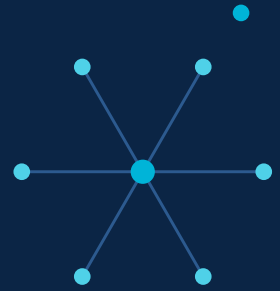


NTI NETWORK SECURITY TRAINING



Practical Labs Portfolio

Routing • DHCP & DHCP Relay • Wireless Networking
VLANs & Trunking • Inter-VLAN Routing

Built & simulated in Cisco Packet Tracer

Three end-to-end labs demonstrating multi-network IP design, centralized DHCP with relay, wireless integration, and VLAN segmentation with Router-on-a-Stick.

Introduction

This document presents a collection of practical networking labs completed during the NTI Network Security Training, built and simulated entirely in Cisco Packet Tracer. Each lab was designed to build progressively on core switching and routing concepts — starting from basic multi-network IPv4 routing, moving through wireless integration and centralized DHCP services, and finishing with VLAN segmentation and inter-VLAN routing using the Router-on-a-Stick model.

For every lab, the topology diagram exported from Packet Tracer is included alongside the addressing plan, the relevant configuration commands, and a short explanation of what the lab demonstrates and why it matters in a real network.

What this portfolio covers

- ▶ Lab 1 — Multi-network routing with static routes and a centralized DHCP server reached via `ip helper-address`.
- ▶ Lab 2 — Extending the network over Wireless Access Points, with DHCP Relay carrying requests across router boundaries.
- ▶ Lab 3 — Segmenting a LAN into VLANs (BA, HR, IT), trunking between switches, and routing between VLANs with Router-on-a-Stick.

Contents

Lab 1 — Multi-Network Routing & Centralized DHCP



5 subnets, inter-router links, DHCP relay

Lab 2 — Wireless Networking & DHCP Relay



Access Points, wireless clients, cross-router DHCP

Lab 3 — VLANs, Trunking & Inter-VLAN Routing



VLAN 10/20/30, 802.1Q trunks, Router-on-a-Stick

Conclusion



Key takeaways from the training

LAB 1

Multi-Network Routing & Centralized DHCP

Connecting five subnets with static routes and a single, centralized DHCP server

1.1 Lab Overview

This lab focuses on connecting multiple networks using Cisco routers and switches. It demonstrates IPv4 addressing across five separate subnets, router-to-router connectivity over point-to-point links, a single centralized DHCP server, and DHCP Relay using the `ip helper-address` command so that clients on remote subnets can still obtain an address from that one server.

1.2 Topology

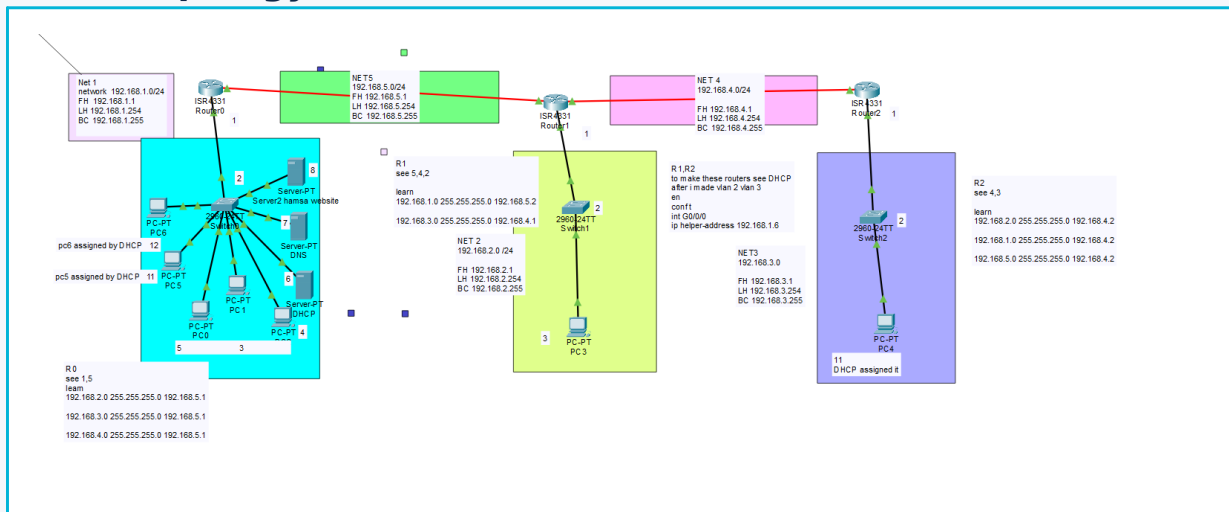


Figure 1.1 — Three routers (R0, R1, R2) interconnect five networks. NET 1 hosts the DHCP server, web server, and DNS server; PCs on NET 1, NET 2, and NET 4 obtain addresses from that same server.

1.3 Addressing Plan

Network	Subnet	First Host	Last Host	Broadcast
NET 1	192.168.1.0/24	192.168.1.1	192.168.1.254	192.168.1.255
NET 2	192.168.2.0/24	192.168.2.1	192.168.2.254	192.168.2.255
NET 3	192.168.3.0/24	192.168.3.1	192.168.3.254	192.168.3.255
NET 4	192.168.4.0/24	192.168.4.1	192.168.4.254	192.168.4.255
NET 5	192.168.5.0/24	192.168.5.1	192.168.5.254	192.168.5.255

1.4 Centralized DHCP & DHCP Relay

The DHCP server sits on NET 1 alongside a web server and a DNS server. PCs directly on NET 1 (PC5, PC6) can reach it directly through broadcast. Clients on NET 2 and NET 4,

however, sit behind a router, and DHCP discovery messages are broadcasts that routers do not forward by default. The fix is DHCP Relay: the router's LAN-facing interface is configured with `ip helper-address 192.168.1.6`, which tells the router to convert the broadcast into a unicast packet aimed directly at the DHCP server, then relay the server's reply back to the client.

1.5 Key Configuration

Enabling DHCP Relay on the router interfaces facing NET 2 / NET 4:

```
R1(config)# interface g0/0/0
R1(config-if)# ip helper-address 192.168.1.6
```

Static routes learned on R0 so it knows how to reach the remote subnets:

```
R0(config)# ip route 192.168.2.0 255.255.255.0 192.168.5.1
R0(config)# ip route 192.168.3.0 255.255.255.0 192.168.5.1
R0(config)# ip route 192.168.4.0 255.255.255.0 192.168.5.1
```

Result

- ▶ PC5 and PC6 on NET 1 receive their IP addresses directly from the local DHCP server.
- ▶ PC4 on NET 4, across two router hops, still receives an address automatically thanks to the relay configuration.
- ▶ All five subnets can reach one another once static routes are in place on R0, R1, and R2.

LAB 2

Wireless Networking & DHCP Relay

Extending the wired network with Access Points and cross-router DHCP

2.1 Lab Overview

This lab focuses on integrating wired and wireless networks using Wireless Access Points. It demonstrates wireless connectivity for laptops, a centralized DHCP server shared across networks, DHCP Relay across a router boundary, and end-to-end communication between wired and wireless devices on different subnets.

2.2 Topology

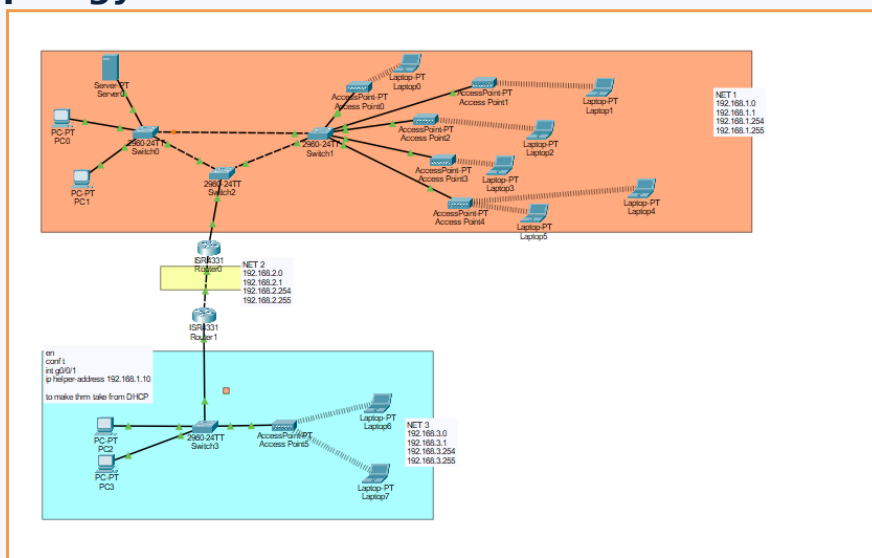


Figure 2.1 — NET 1 (orange) hosts the DHCP server and several Access Points serving wireless laptops; NET 3 (blue) is a separate site reached through Router0 and Router1, with its own Access Point and wired PCs.

2.3 Wireless Coverage



Figure 2.2 — Simulated coverage footprint of the Access Points across the floor plan, showing signal overlap between rooms.

2.4 Addressing Plan

Network	Subnet	Gateway	Last Host	Broadcast
NET 1	192.168.1.0/24	192.168.1.1	192.168.1.254	192.168.1.255
NET 2	192.168.2.0/24	192.168.2.1	192.168.2.254	192.168.2.255
NET 3	192.168.3.0/24	192.168.3.1	192.168.3.254	192.168.3.255

2.5 Access Points & DHCP Relay

Each Access Point acts as a Layer-2 bridge between the wireless laptops and the wired switch it is plugged into — laptops associate with an AP using SSID and security settings, then behave exactly like any wired host on that subnet. Because NET 3 sits behind Router1, its DHCP requests are relayed to the central DHCP server on NET 1 using the same `ip helper-address` technique from Lab 1.

2.6 Key Configuration

On Router1's interface facing NET 3, so its clients pull addresses from the DHCP server on NET 1:

```
Router1(config)# interface g0/0/1
Router1(config-if)# ip helper-address 192.168.1.10
```

Result

- ▶ Laptops connected to Access Points 0–4 on NET 1 associate wirelessly and obtain addresses from the local DHCP server.
- ▶ PC2, PC3, and Laptop6/7 on the remote NET 3 site obtain addresses automatically via DHCP Relay, despite being two router hops away.
- ▶ Wired and wireless clients across all three networks can ping one another once routing is in place.

LAB 3

VLANs, Trunking & Inter-VLAN Routing

Segmenting one switched network into three virtual LANs with Router-on-a-Stick

3.1 Lab Overview

This lab focuses on network segmentation using VLANs and trunk links. It demonstrates the configuration of three VLANs — VLAN 10 (BA), VLAN 20 (HR), and VLAN 30 (IT) — spread across three access switches, trunked back to a distribution switch, with a router performing Router-on-a-Stick inter-VLAN routing over a single physical link using 802.1Q sub-interfaces.

3.2 Topology

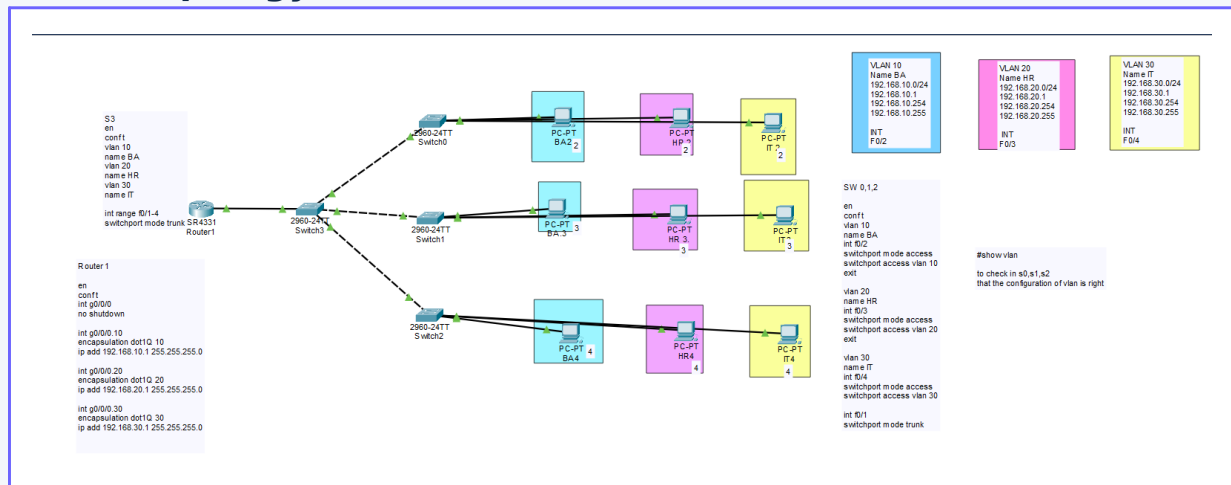


Figure 3.1 — Switch0, Switch1, and Switch2 each carry all three VLANs to their access ports; Switch3 trunks everything up to Router1, which routes between VLANs.

3.3 VLAN & Addressing Plan

VLAN 10 • BA		VLAN 20 • HR		VLAN 30 • IT	
Subnet	192.168.10.0/24	Subnet	192.168.20.0/24	Subnet	192.168.30.0/24
Gateway	192.168.10.1	Gateway	192.168.20.1	Gateway	192.168.30.1
Last host	192.168.10.254	Last host	192.168.20.254	Last host	192.168.30.254
Broadcast	192.168.10.255	Broadcast	192.168.20.255	Broadcast	192.168.30.255
Access port	Fa0/2	Access port	Fa0/3	Access port	Fa0/4

3.4 Switch Configuration

Creating the VLANs and naming them on each access switch (repeated on Switch0, 1, 2):

```
Switch(config)# vlan 10
Switch(config-vlan)# name BA
Switch(config)# vlan 20
Switch(config-vlan)# name HR
Switch(config)# vlan 30
Switch(config-vlan)# name IT
```

Assigning access ports to their VLAN:

```
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config)# interface fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
```

Trunking the uplinks between switches, and from Switch3 up to the router:

```
Switch3(config)# interface range fa0/1-4
Switch3(config-if-range)# switchport mode trunk
```

3.5 Router-on-a-Stick Configuration

A single physical link from Router1 to Switch3 carries traffic for all three VLANs, tagged with 802.1Q. The router uses one sub-interface per VLAN, each acting as that VLAN's default gateway:

```
Router1(config)# interface g0/0/0
Router1(config-if)# no shutdown
Router1(config)# interface g0/0/0.10
Router1(config-subif)# encapsulation dot1q 10
Router1(config-subif)# ip address 192.168.10.1 255.255.255.0
Router1(config)# interface g0/0/0.20
Router1(config-subif)# encapsulation dot1q 20
Router1(config-subif)# ip address 192.168.20.1 255.255.255.0
Router1(config)# interface g0/0/0.30
Router1(config-subif)# encapsulation dot1q 30
Router1(config-subif)# ip address 192.168.30.1 255.255.255.0
```

Running `show vlan` on each switch confirms the correct ports are assigned to each VLAN before testing end-to-end connectivity.

Result

- ▶ Hosts within the same VLAN (e.g. all BA machines across the three switches) communicate directly at Layer 2.
- ▶ Hosts in different VLANs (e.g. BA to HR) are routed through Router1's sub-interfaces — true inter-VLAN routing.
- ▶ Broadcast traffic is contained within each VLAN, reducing broadcast domain size and improving security segmentation.

4 Conclusion

These three practical labs provided hands-on experience with core Cisco networking technologies and strengthened my understanding of end-to-end network design. Across the labs I worked with:

- ▶ IPv4 addressing and static routing across multiple interconnected subnets.
- ▶ Centralized DHCP services and DHCP Relay, allowing one server to address clients on remote subnets.
- ▶ Wireless integration, connecting Access Points and laptops into an existing wired topology.
- ▶ VLAN segmentation and trunking, isolating departments (BA, HR, IT) at Layer 2.
- ▶ Router-on-a-Stick inter-VLAN routing, enabling controlled communication between VLANs over a single trunk link.

Together, these labs mirror the building blocks of a small-to-medium enterprise network and reinforced how routing, DHCP, wireless access, and VLAN segmentation work together to deliver a scalable and secure network design.

✓ **All three labs were built, configured, and verified end-to-end in Cisco Packet Tracer.**